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Lab Report: RF Components: Characterization and Measurement

Experiments performed (Part 1): Filter Characterization (A1), Filter Return Loss (A2), Quarter-wave Stub Notch Filter (A3), Mixer Characterization (Part B), Power Splitter/Combiner Characterization (Part C)

1. Objective

The objective of this experiment is to understand the operation of, and experimentally characterize, the passive and active RF components used in a typical RF receiver chain. Specifically, this report covers:

- To characterize an RF filter, identify its type from its measured transmission response, and determine its cutoff frequency.
- To estimate the input reflection coefficient and return loss of the filter.
- To demonstrate a quarter-wave open-circuit stub as a notch filter and compare the calculated and measured notch frequency.
- To observe frequency conversion in an RF mixer and estimate its conversion loss.
- To measure the power division and insertion loss of a power splitter and to characterize a power combiner.

2. Theory

2.1 RF Components in an RF Receiver Chain

A minimal RF receiver chain consists of a low-noise amplifier (LNA), a mixer, filters, and a power splitter/combiner, followed by an oscilloscope or spectrum analyzer. A filter selects or rejects a specified range of frequencies. A mixer translates a signal from one frequency to another so that our system can utilize that signal. A power splitter/combiner distributes a signal to multiple paths or combines signals arriving from different paths.

2.2 Filter Transmission Characteristics

A filter passes selected frequency components and attenuates others. The four common filter types are the low-pass filter (passes frequencies below a cutoff f_c), the high-pass filter (passes frequencies above f_c), the band-pass filter (passes frequencies between two cutoffs f_1 and f_2), and the band-reject/notch filter (attenuates a specified stopband).

For a transmission measurement, if P_{in} is the power measured without the filter in the chain and P_{out} is the power measured with the filter inserted, the insertion loss is defined as:

$$IL = P_{in} - P_{out} \text{ (dB)}$$

A smaller IL indicates better transmission.

2.3 Filter Reflection and Return Loss

When the filter input impedance Z_{in} is not equal to the system characteristic impedance Z_0, part of the incident signal is reflected. The voltage reflection coefficient is:

$$\Gamma = V_{ref} / V_{inc} = (Z_{in} - Z_0) / (Z_{in} + Z_0)$$

Using the Tee-connector method, the maximum and minimum standing-wave voltage amplitudes (V_{max}, V_{min}) observed while sweeping frequency near the Tee junction give:

$$V_{in} = (V_{max} + V_{min})/2,\ V_{ref} = (V_{max} - V_{min})/2\ ,\ |\Gamma| = V_{ref}/V_{inc}$$

The return loss is then:

$$RL = -20 \log10 |\Gamma| \text{ (dB)}$$

A perfectly matched filter has $|\Gamma| = 0$ and $RL \to \infty$, while complete reflection gives $|\Gamma| = 1$ and RL = 0 dB.

2.4 Quarter-wave Open-circuit Stub Filter

An open-ended coaxial stub connected in shunt through a Tee connector acts as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end transforms into a low impedance at the Tee junction, strongly attenuating the signal and producing a notch. The physical stub length required for a notch at frequency f_0 is:

$$L = \lambda/4$$

And the notch frequency f_0 is defined as:

$$f_o = VF \cdot c / (4L)$$

where VF is the velocity factor of the cable and c is the speed of light in vacuum. Connector length and open-end effects can cause the measured notch frequency to differ slightly from the calculated value.

2.5 Mixer

An RF mixer is a three-port nonlinear device with RF, LO, and IF ports. When signals at f_{RF} and f_{LO} are applied, sum and difference frequency components are generated at the IF port:

$$f_{IF} = f_{RF} + f_{LO} \text{ and } f_{IF} = f_{RF} - f_{LO}$$

If P_{RF} is the RF input power and P_{IF} is the measured power of the selected IF component, the conversion loss is:

$$\text{Conversion loss} = P_{RF} - P_{IF} \text{ (dB)}$$

2.6 Power Splitter/Combiner

A power splitter divides input power between two or more output ports; for an ideal two-way equal splitter, each output is 3 dB below the input. For input power P_{in} and output powers P_{out1} and P_{out2}, the losses to each port are:

$$Loss_1 = P_{in} - P_{out1} \text{ and } Loss_2 = P_{in} - P_{out2}$$

A power combiner performs the reverse operation; its output depends on the frequency, amplitude, and relative phase of the input signals.

3. Components Required

- RF signal generator(s)
- Spectrum analyser
- Digital storage oscilloscope
- Filter (device under test)
- Mixer
- Power splitter/combiner
- RF Tee connector
- Open-ended coaxial stub
- Short reference coaxial cable
- 50 Ω terminations
- Suitable RF cables, adapters, and attenuators

4. Experimental Setup

4.1 Setup A - Transmission Measurements

Used for the filter insertion-loss measurement (A1), the stub notch-filter measurement (A3), and the mixer measurement (Part B). The RF signal generator is connected to the device under test (DUT), and the output is measured with the spectrum analyser. A reference measurement (without the DUT) is first taken with a short cable at the same generator power, and then the DUT is inserted without changing the generator setting, cables, or adapters.

4.2 Setup B - Filter Return-loss Measurement

Used for the return-loss measurement (A2). The RF signal generator is connected to one port of a Tee connector, the oscilloscope (set to 1 MΩ input impedance) to the second port, and the filter input to the third port. The filter output is terminated with a 50 Ω load. The generator amplitude is held constant while the frequency is swept slowly across a flat part of the passband, and the maximum and minimum voltage amplitudes at the Tee junction are recorded.

4.3 Splitter/Combiner Setup (Part C)

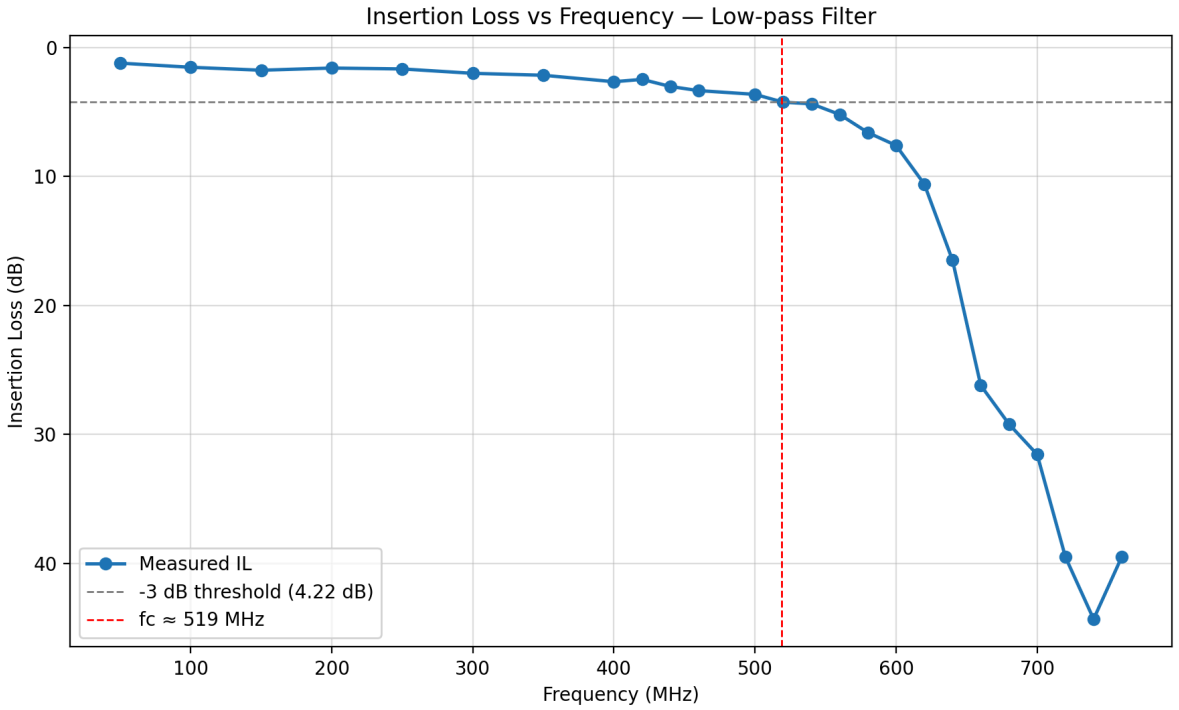
For the splitter, the input is connected to the signal generator, one output port to the spectrum analyser, and the unused output port terminated with 50 Ω; the connections are then interchanged to measure the second output. For the combiner, two signal generators are connected to the two input ports, and the combined output is measured on the spectrum analyser.

5. Observations and Graphs

5.1 Part A1 - Filter Transmission Response and Insertion Loss

Input Frequency (MHz)	Pin without filter (dBm)	Pout with filter (dBm)	IL (dB)
50	0.00	-1.22	1.22
100	0.00	-1.54	1.54
150	0.00	-1.77	1.77
200	0.00	-1.60	1.60
250	0.00	-1.67	1.67
300	0.00	-2.01	2.01
350	0.00	-2.16	2.16
400	0.00	-2.66	2.66
420	0.00	-2.48	2.48
440	0.00	-3.03	3.03
460	0.00	-3.35	3.35
500	0.00	-3.64	3.64
520	0.00	-4.25	4.25
540	0.00	-4.37	4.37
560	0.00	-5.20	5.20
580	0.00	-6.60	6.60
600	0.00	-7.60	7.60
620	0.00	-10.60	10.60
640	0.00	-16.50	16.50
660	0.00	-26.20	26.20
680	0.00	-29.20	29.20
700	0.00	-31.56	31.56

720	0.00	-39.50	39.50
740	0.00	-44.32	44.32
760	0.00	-39.48	39.48



5.2 Part A2 - Filter Input Reflection Coefficient and Return Loss

The frequency was swept slowly across the passband while observing the standing-wave voltage at the Tee junction. The maximum voltage amplitude was recorded at 268 MHz and the minimum at 220 MHz:

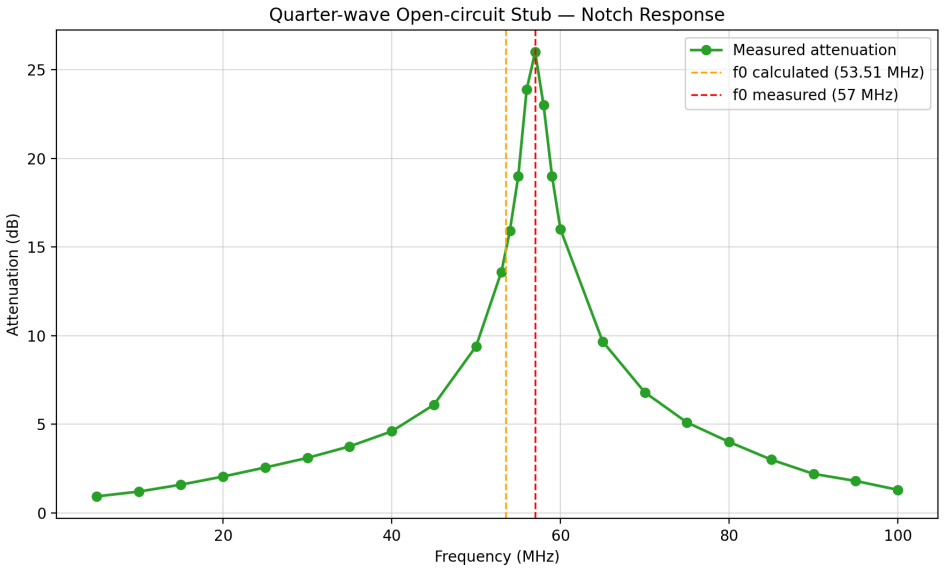
Frequency (MHz)	Vmax (a.u.)	Vmin (a.u.)
268 (Vmax point)	155.77	-
220 (Vmin point)	-	129.75

5.3 Part A3 - Quarter-wave Open-circuit Stub Notch Filter

Stub parameters: velocity factor VF = 0.66, stub length L = 92.5 cm.

Frequency (MHz)	Pin without stub (dBm)	Pout with stub (dBm)	Attenuation (dB)

5	0.00	-0.93	0.93
10	0.00	-1.20	1.20
15	0.00	-1.59	1.59
20	0.00	-2.05	2.05
25	0.00	-2.56	2.56
30	0.00	-3.10	3.10
35	0.00	-3.75	3.75
40	0.00	-4.60	4.60
45	0.00	-6.10	6.10
50	0.00	-9.40	9.40
53	0.00	-13.60	13.60
54	0.00	-15.90	15.90
55	0.00	-19.00	19.00
56	0.00	-23.90	23.90
57	0.00	-26.00	26.00
58	0.00	-23.00	23.00
59	0.00	-19.00	19.00
60	0.00	-16.00	16.00
65	0.00	-9.67	9.67
70	0.00	-6.80	6.80
75	0.00	-5.10	5.10
80	0.00	-4.00	4.00
85	0.00	-3.00	3.00
90	0.00	-2.20	2.20
95	0.00	-1.80	1.80
100	0.00	-1.30	1.30



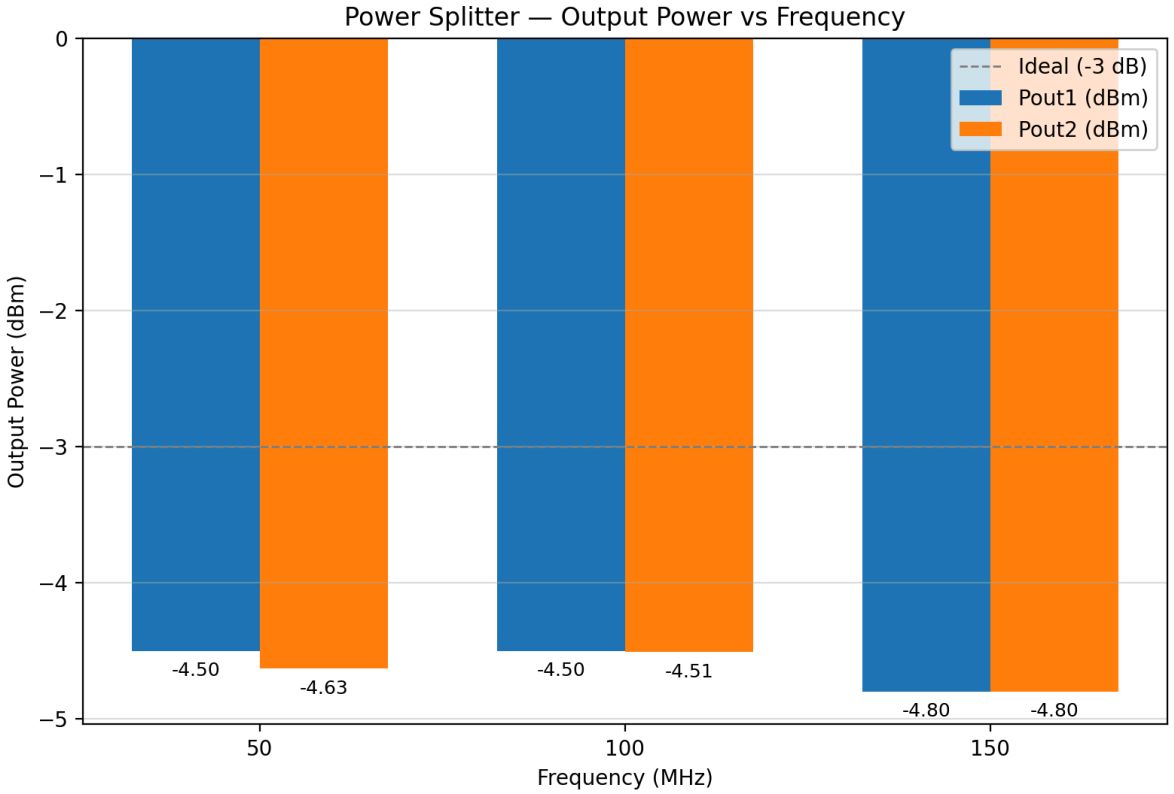
5.4 Part B - Mixer Characterization

Supplied RF power P_{RF} = −3.00 dBm, LO power P_{LO} = 0.00 dBm.

f_{RF} (MHz)	f_{LO} (MHz)	Expected f_{IF} (MHz)	Measured f_{IF} (MHz)	P_{IF} (dBm)	Conversion loss (dB)
30	10	20	19.99	-13.35	10.35
30	10	40	39.99	-13.37	10.37
40	10	50	49.97	-13.30	10.30
40	10	30	29.97	-13.50	10.50
50	10	60	59.99	-13.30	10.30
50	10	40	39.99	-13.20	10.20

5.5 Part C1 - Power Splitter Characterization

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	$Loss_1$ (dB)	$Loss_2$ (dB)
50	0.00	-4.50	-4.63	4.50	4.63
100	0.00	-4.50	-4.51	4.50	4.51
150	0.00	-4.80	-4.80	4.80	4.80



5.6 Part C2 - Power Combiner Characterization

Both signal generators were set to equal power (−3 dBm each) at the same frequency and combined.

Frequency (MHz)	P_1 (dBm)	P_2 (dBm)	Ideal dB-sum P_1+P_2 (dBm)	Measured P_{out} (dBm)	Excess loss (dB)
50	-3.00	-3.00	-6.00	-7.26	1.26
100	-3.00	-3.00	-6.00	-7.41	1.41
150	-3.00	-3.00	-6.00	-7.54	1.54

6. Calculations and Results

6.1 Filter Type Identification and Cutoff Frequency

From the A1 data, the insertion loss is small (≈1.2–1.7 dB) and roughly flat for input frequencies from 50–250 MHz, then increases gradually up to ≈460 MHz, and rises sharply beyond ≈600 MHz, reaching over 40 dB by 720 MHz. This monotonically increasing attenuation with frequency, passing low frequencies and rejecting high frequencies, is the signature of a low-pass filter. No second passband edge or symmetric roll-off is observed, ruling out a band-pass or notch response.

The minimum insertion loss in the passband is IL(min) = 1.22 dB (at 50 MHz). The −3 dB cutoff is therefore defined at IL = 1.22 + 3 = 4.22 dB. Interpolating linearly between the measured points at 500 MHz (IL = 3.64 dB) and 520 MHz (IL = 4.25 dB):

$$fc \approx 500 + [(4.22 - 3.64)/(4.25 - 3.64)] \times 20 \approx 519 \text{ MHz}$$

Hence, the filter is identified as a low-pass filter with a passband from DC to approximately 519 MHz. The sharp roll-off above 600 MHz (IL > 10 dB by 620 MHz, > 30 dB by 700 MHz) confirms a well-defined stopband. The slight dip in IL at 760 MHz (39.48 dB, less attenuation than at 740 MHz) is most likely a spurious/ripple response or a measurement artifact near the noise floor of the spectrum analyser.

6.2 Filter Reflection Coefficient and Return Loss

Using V_{max} = 155.77V and V_{min} = 129.75V:

$$V_{inc} = (155.77 + 129.75)/2 = 142.76$$

$$V_{ref} = (155.77 - 129.75)/2 = 13.01$$

$$|\Gamma| = 13.01 / 142.76 = 0.0911$$

$$RL = -20 \log10(0.0911) \approx 20.8 \text{ dB}$$

A return loss of ≈20.8 dB corresponds to a fairly well-matched filter ($|\Gamma| \approx 0.09$, i.e. only ~0.8% (i.e., $|\Gamma|^2$~0.0083 or 0.8%) of incident power reflected), though not a perfect match. This is consistent with a practical filter having some residual impedance mismatch at its input port.

6.3 Quarter-wave Stub Notch Frequency

The calculated notch frequency, using VF = 0.66 and L = 92.5 cm:

$$f_0(\text{theoretical}) = (VF \cdot c)/(4L) = (0.66 \times 3\times10^8)/(4 \times 0.925) \approx 53.51 \text{ MHz}$$

The measured data shows the deepest attenuation (26 dB) at 57 MHz, which is taken as the measured notch frequency:

$$f_0(\text{measured}) = 57 \text{ MHz}$$

The percentage difference between calculated and measured notch frequency is:

$$\% \text{ difference} = |57 - 53.51| / 53.51 \times 100 \approx 6.5\%$$

This small discrepancy is attributable to the connector length and open-end fringing capacitance at the stub tip, both of which effectively add to the electrical length of the stub and are not accounted for in the simple λ/4 formula, as noted in the manual.

6.4 Mixer Conversion Loss

For each RF/LO combination, the conversion loss was calculated as $CL = P_{RF} - P_{IF}$, with P_{RF} = −3.00 dBm fixed:

- f_{RF}=30, f_{LO}=10 → f_{IF}=20 MHz: CL = −3.00 − (−13.35) = 10.35 dB
- f_{RF}=30, f_{LO}=10 → f_{IF}=40 MHz: CL = −3.00 − (−13.37) = 10.37 dB

- f_{RF}=40, f_{LO}=10 → f_{IF}=50 MHz: CL = −3.00 − (−13.30) = 10.30 dB
- f_{RF}=40, f_{LO}=10 → f_{IF}=30 MHz: CL = −3.00 − (−13.50) = 10.50 dB
- f_{RF}=50, f_{LO}=10 → f_{IF}=60 MHz: CL = −3.00 − (−13.30) = 10.30 dB
- f_{RF}=50, f_{LO}=10 → f_{IF}=40 MHz: CL = −3.00 − (−13.20) = 10.20 dB

The measured f_{IF} in every case matches the expected $f_{RF} \pm f_{LO}$ value to within 0.03 MHz, confirming correct mixer operation. The conversion loss is consistent across all six combinations, averaging ≈10.34 dB, indicating a stable and roughly frequency-independent conversion loss over this range.

6.5 Power Splitter - Division and Insertion Loss

The measured losses to each output port (relative to P_{in} = 0 dBm) range from 4.50–4.80 dB, compared to the ideal equal-split value of 3 dB. This indicates an additional ≈1.5–1.8 dB of excess insertion loss beyond the ideal 3 dB power division, arising from cable, connector, and splitter dissipative losses.

As a cross-check, a separate voltage-ratio measurement gave an output-to-input voltage ratio of 609.3 mV / 1 V = 0.6093 (compared to the ideal $1/\sqrt{2} = 0.707$ for a lossless 3 dB split). Since $P \propto V^2$, this voltage ratio corresponds to a loss of:

$$\text{Loss} = -20 \log_{10}(0.6093) \approx 4.30 \text{ dB}$$

which agrees closely with the ≈4.5–4.8 dB directly measured on the spectrum analyser, confirming the $P \propto V^2$ relationship and validating the power-based loss measurement.

6.6 Power Combiner

With both input tones at equal power (−3 dBm each, same frequency), the dB-sum expectation is P1 + P2 = −3 + (−3) = −6 dBm. The measured combined output power was −7.26, −7.41, and −7.54 dBm at 50, 100, and 150 MHz, respectively, an excess loss of 1.26–1.54 dB beyond the −6 dBm estimate. This additional loss represents the combiner's own insertion/dissipative loss, analogous to the splitter case (the combiner being the reciprocal device to the splitter).

7. Conclusion

Summary of key results:

- Filter identified as low-pass, with a −3 dB cutoff frequency fc ≈ 519 MHz.
- Filter return loss RL ≈ 20.8 dB (|Γ| ≈ 0.091).
- Quarter-wave stub notch filter: calculated f_o = 53.51 MHz, measured f_o = 57 MHz (≈6.5% difference).
- Mixer conversion loss ≈ 10.3 dB, consistent across all tested f_{RF}/f_{LO} combinations.
- Power splitter: measured insertion loss ≈ 4.5–4.8 dB per output vs. ideal 3 dB.
- Power combiner: measured output ≈1.3–1.5 dB below the naive dB-sum of the two input powers.

7. Key Learnings

This experiment gave me a working understanding of a basic RF receiver chain and how each block in it, the filter, mixer, splitter/combiner, and the spectrum analyzer actually behaves when measured. I learnt how a spectrum analyzer is used to read out power levels at different frequencies and how that lets you build up the transmission response of a filter and identify its type just from the shape of the data. I also learnt the difference between insertion loss and return loss. Insertion loss tells you how much power actually gets through a device, while return loss tells you how well matched the device is at its input and how much voltage has been reflected back instead of being transmitted forward. On the return loss side, I understood why matching the impedance of the device to the system impedance is what minimizes reflection, and why a mismatch sends part of the signal back instead of letting it pass through. I also learnt that power is proportional to V^2, so when power is split equally between two paths, the voltage in each path is scaled by $1/\sqrt{2}$ and not by 1/2, which I confirmed by comparing my power-based and voltage-based loss measurements on the splitter. Lastly, working with the actual splitter and combiner showed me that real components never match the ideal numbers exactly. The splitter lost close to 4.5 to 4.8 dB per output instead of the ideal 3 dB, and the combiner output was about 1.3 to 1.5 dB lower than the simple dB sum of the two inputs, which showed me that every real device adds its own extra loss on top of what the theory predicts.